

Introduction

- **Question:** Can environmental conditions predict whether a fishing trip will be successful?
- Focus on trout fishing in Idaho or California, but I am considering other species and regions depending on the availability of usable data. I plan to select a specific region rather than study the entire United States.
- I will investigate variables such as water temperature, streamflow, water level, precipitation, month, time of day, location, elevation, and fish species.
- A central part of the project will be combining fishing records with environmental data from separate sources. I am still identifying additional datasets

Background

- Research on why environmental conditions are relevant to fish habitat and behavior.
- Explain the distinction between fish abundance, the number of fish caught, and catch rate
 - I want an outcome that accounts for fishing effort rather than simply counting fish.
- Discuss how this project will move from ecological background research to testing whether environmental information can help predict observed fishing outcomes.

Questions This Data Can Answer

- Can environmental conditions predict whether catch rate exceeds a chosen definition of success?
- Which environmental variables provide the most useful information for predicting fishing success?
- Do catch rates and their relationships with environmental conditions differ across seasons, locations, or species, where sufficient observations are available?
- Does adding water and weather information improve predictions compared with a simpler model that uses only location and season?

Data and Key Variables

- I will prioritize angler or creel-survey records containing fishing dates, locations, fish caught, species, and fishing effort.
 - Still need to inspect the downloadable files to establish which fields and observation types are available for my analysis.

Environmental data

- USGS Water Data for the Nation: Investigate water-monitoring records at stations relevant to the selected fishing locations, checking the availability of streamflow, water-level, and water-temperature measurements during the study period.
- NOAA Climate Data Online: Obtain historical weather measurements, particularly daily precipitation and air temperature, for the selected region and dates.

Getting The Data

- Download the fishing data and accompanying documentation from the selected agency or repository. Read the documentation to identify variable definitions, units, sampling methods, and whether catches include released fish or only harvested fish.
- Identify relevant water-monitoring and weather stations, then obtain environmental records covering the same dates as the fishing data.
- Check geographic and temporal overlap before committing to a dataset combination.
- Keep a record of source names, download dates, study periods, and access procedures so that the data collection can be repeated.

Cleaning, Merging, and Analysis

Cleaning and preparing the data

- Standardize dates, location names, species labels, and measurement units. Check for duplicate records, missing values, and invalid fishing durations.
- Retain recorded trips with zero catches, but do not treat dates without a fishing observation as zero-catch days.
- Calculate catch rate and the success label. Keep the catch information used to create the outcome separate from the environmental predictors.

Merging the datasets

- Create a matching table connecting each fishing location to an appropriate water-monitoring station and weather station.
- Merge records using the station assignments and corresponding dates. Match the environmental summaries to the fishing data's time scale
- Check how many records successfully match, investigate unmatched observations, and confirm that the joins have not accidentally duplicated fishing records.

Exploration and modeling

- Produce graphs showing catch-rate patterns across environmental conditions, seasons, and locations.
- Consider a simple classification model, such as logistic regression, and compare it with a random forest if the dataset is suitable.
- Report both successful and unsuccessful predictions rather than relying only on overall accuracy.

Ethical Considerations

- Use publicly available data according to each source's terms, credit the organizations that collected them, and exclude personal identifiers from the analysis.
- Consider whether publishing precise fishing locations could place additional pressure on sensitive fisheries; present results at a broader geographic scale where appropriate.
- Describe predictions as uncertain estimates rather than guarantees.

Limitations

- Unmeasured influences: I may not have information about angler experience, fishing method, bait, stocking, or other factors that could help explain differences in catch rates.

- Sampling and reporting: I will examine whether the records disproportionately represent certain locations, seasons, types of anglers, or kinds of catches.
- Environmental matching: A monitoring station may not capture the exact conditions at the fishing location.

Expected Outcome and Next Steps

- A cleaned, merged dataset linking fishing observations with environmental conditions, accompanied by a record of the cleaning and matching decisions.
- Visualizations and a comparison of predictive models showing whether environmental variables improve predictions.
- An interpretation of the findings, including the possibility that the available environmental information does not substantially improve prediction.

References

Continue developing the existing annotated bibliography, distinguishing background research from datasets actually used.